

Gallium alloys are ideal candidates for flexible electronics as these liquid metals are malleable and conductive. Also, the researchers could manipulate gallium alloys easily without touching them. For liquid metal based flexible printed electronics, the ideal properties of the core manufacturing material, i.e., room-temperature liquid metal—currently mainly represented by gallium and its alloys—are excellent resistivity, enormous bendability, low adhesion and large surface tension. With such properties, any droplet of liquid metal contains a highly-conductive metallic core and an atomically thin, semiconducting oxide skin. These are the key parameters for developing flexible electronic circuits.

The suitability of gallium for future electronics was demonstrated recently by researchers in a study in which they (researchers) immersed

liquid metal droplets in water. Due to ionic interactions, the metal was able to move freely in three dimensions—the key to developing a suitable material in understanding the influence of the surrounding fluid on the metal. The researchers showed how adjustments made to concentrations of acid, base and salt components in the water were influential. Getting this right should open the door to creating moving objects, switches and pumps—each necessary for flexible electronic devices.

Technological developments

Scientists are concerned that the fundamental technology of electronics has not progressed much for many decades, and mobile phones and computers today are no more powerful than they were five years ago. They are searching for a technology that will allow them to increase the processing power of

electronic components.

Researchers believe that the answer lies in ultra-thin components made by printing with liquid metals. The 2D printing technique can create many layers of incredibly thin electronic chips on the same surface, dramatically increasing the processing power while reducing costs. Using liquid metals to create integrated circuits that are just atoms thick could lead to the next big advance in electronics. The process opens the way for production of large wafers around 1.5 nanometres (nm) in depth; a sheet of paper, by comparison, is 100,000nm thick. Other techniques have proven unreliable in terms of quality, scaling and functioning at very high temperatures (550 degrees or higher). Researchers hope that creating electronic wafers just atoms thick could overcome the limitations of current chip production. It could



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